

Dr. Luwei Yang

Physical Oceanographer

Department of Ocean Science

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Research Profile

My research addresses a fundamental challenge in physical oceanography: understanding how energy is transferred across scales, from basin-scale ocean circulation to small-scale turbulent mixing, and representing these transfers in global ocean models. I focus particularly on internal waves, which transfer energy across this wide range of scales but remain unresolved in global ocean and climate models. My work combines theory, numerical modelling, and observations to investigate internal-wave energy pathways and develop physically based parameterisations that bring their effects into global ocean and climate models.

Academic Appointments

- **Postdoctoral Research Fellow** 2026–present
Department of Ocean Science, The Hong Kong University of Science and Technology
Hong Kong, China
- **Visiting and Honorary Appointee** 2026–present
Research School of Earth Sciences, Australian National University
Canberra, ACT, Australia
- **Postdoctoral Research Fellow** 2023–2026
Research School of Earth Sciences, Australian National University
Canberra, ACT, Australia
- **Postdoctoral Researcher** 2020–2023
Department of Atmospheric & Oceanic Sciences, University of California, Los Angeles
Los Angeles, California, United States

Education

- **PhD in Quantitative Marine Science** 2015–2019
University of Tasmania
Hobart, Tasmania, Australia
- **BSc in Marine Science** 2011–2015
Ocean University of China
Qingdao, China

Publications

Peer-reviewed journal articles

- [8] **Yang, L.**, Shakespeare, C. J., and B.K. Arbic, 2026: Propagating internal-tide-induced wave stresses resolve discrepancies in ocean surface tide energetics. *Geophysical Research Letters*, 53, e2026GL123598, doi: 10.1029/2026GL123598

- [7] **Yang, L.**, C.J. Shakespeare, A.K. Morrison, A.M. Hogg, A.H. Gibson, and B.K. Arbic, 2026: Sensitivity of M_2 barotropic tide solutions to resolution and a physically based wave drag parameterization. *Journal of Advances in Modeling Earth Systems*, 18, e2025MS005528, doi: 10.1029/2025MS005528
- [6] Barnes, A.J., N.C. Constantinou, A.H. Gibson, A.E. Kiss, C. Chapman, J. Reilly, D. Bhagtani, **L. Yang**, 2024: regional-mom6: A Python package for automatic generation of regional configurations for the Modular Ocean Model 6. *Journal of Open Source Software*, **9(100)**, doi: 10.21105/joss.06857
- [5] **Yang, L.**, R. Barkan, K. Srinivasan, J.C. McWilliams, C.J. Shakespeare, and A.H. Gibson, 2023: Oceanic eddies induce a rapid formation of an internal wave continuum. *Communications Earth & Environment*, **4(1)**, doi: 10.1038/s43247-023-01137-1
- [4] **Yang, L.**, M. Nikurashin, A.M. Hogg, and B.M. Sloyan, 2023: Lee waves break eddy saturation of the Antarctic Circumpolar Current. *Geophysical Research Letters*, **50(11)**, doi: 10.1029/2023GL103866
- [3] Barkan R., K. Srinivasan, **L. Yang**, J.C. McWilliams, J. Gula, and C. Vic, 2021: Oceanic mesoscale eddy depletion catalyzed by internal waves. *Geophysical Research Letters*, **48(18)**, doi: 10.1029/2021GL094376
- [2] **Yang, L.**, M. Nikurashin, A.M. Hogg, and B.M. Sloyan, 2021: The impact of lee waves on the Southern Ocean circulation. *Journal of Physical Oceanography*, **51(9)**, 2933-2950, doi: 10.1175/JPO-D-20-0263.1
- [1] **Yang, L.**, M. Nikurashin, A.M. Hogg, and B.M. Sloyan, 2018: Energy Loss from Transient Eddies due to Lee Wave Generation in the Southern Ocean. *Journal of Physical Oceanography*, **48(12)**, 2867–2885, doi: 10.1175/JPO-D-18-0077.1

Other publications

- [1] **Yang, L.**, C.J. Shakespeare, 2026: Earth’s tides are shaped by more than the Moon. And they’re changing as the planet warms. *The Conversation*, doi: 10.64628/AA.9gu6v4jtm

Selected Honours and Awards

- **Best Early Career Researcher Presentation** 2024
Ocean Modelling and Observations Workshop, Canberra, ACT, Australia
- **Competitively Selected Participant (fully funded)** 2021
Physical Oceanography Dissertations Symposium (PODS) XI, Lihue, Hawai’i, United States
- **Student Oral Presentation Honourable Mention** 2017
AMOS/MSNZ Conference & ANZ Climate Forum, Canberra, ACT, Australia
- **National Scholarship (¥8,000)** 2014
Ministry of Education of China

Presentations

Invited Talks

- **Propagating internal-tide-induced wave stresses resolve discrepancies in ocean tide energetics**
Gordon Research Seminar on Ocean Mixing, South Hadley, Massachusetts, United States
Registration supported by the conference organisers. June 2026

- **The impact of lee waves on the Southern Ocean circulation and its response to changes in wind stress**
Departmental Seminar, Department of Ocean Science, The Hong Kong University of Science and Technology, online September 2022
- **The impact of lee waves on the Southern Ocean circulation and its response to changes in wind stress**
Physical Oceanography Dissertations Symposium (PODS) XI, Lihue, Kaua'i, Hawai'i, United States
Competitively selected; participation fully funded by the symposium sponsors. October 2021
- **The impact of lee waves on the Southern Ocean circulation and its response to changes in wind stress**
Polar Oceans Seminar, British Antarctic Survey, online November 2020

Selected Oral Presentations

- **Implementing and evaluating a novel tidal wave-drag parameterization in MOM6** 2024
Ocean Modelling and Observations Workshop, Canberra, ACT, Australia
Awarded Best Early Career Researcher Presentation.
- **Oceanic eddies induce a rapid formation of an internal-wave continuum** 2024
CASPO Seminar, Scripps Institution of Oceanography, La Jolla, California, United States
- **Lee waves break eddy saturation of the Antarctic Circumpolar Current** 2024
Ocean Sciences Meeting 2024, HE41B-07, New Orleans, Louisiana, United States
- **Oceanic eddies induce a rapid formation of an internal-wave continuum** 2024
Ocean Sciences Meeting 2024, PS21A-07, New Orleans, Louisiana, United States
- **Impacts of lee waves on the Southern Ocean circulation and its sensitivity to wind stress** 2019
22nd Conference on Atmospheric and Oceanic Fluid Dynamics, Idealized Model Approaches to the Atmosphere and Ocean Circulation 7.2, Portland, Maine, United States
- **The role of lee waves in the dissipation of transient eddies in the Southern Ocean** 2018
Ocean Sciences Meeting 2018, PO12A-04, Portland, Oregon, United States
- **The role of lee waves in the dissipation of transient eddies in the Southern Ocean** 2017
21st Conference on Atmospheric and Oceanic Fluid Dynamics and 19th Conference on Middle Atmosphere, Mesoscale Ocean Dynamics 13.2, Portland, Oregon, United States
- **The role of lee waves in the dissipation of transient eddies in the Southern Ocean** 2017
AMOS/MSNZ Conference and ANZ Climate Forum, Canberra, ACT, Australia
Student Oral Presentation Honourable Mention.

Teaching Experience

- **Lecturer: Fundamentals of Climate Science (EMSC2021)** Semester 2, 2024
Australian National University
Core second-year course in the Research School of Earth Sciences.
Enrolments: 45 undergraduate and master's students.
 - 2 weeks, co-taught within a flipped-classroom format.
 - Delivered lectures (1-hour) and facilitated problem-solving tutorials (1-hour) and computational laboratories (3-hour), marked assignments and exam papers.
 - Received Student Experience of Learning and Teaching (SELT) ratings for respect towards students (4.8/5), stimulating interest (4.5/5), and supporting student learning (4.5/5).

Research Supervision

- **PhD Co-supervisor: Anupiya Ellepola** 2024–present
Australian National University
 - Project: *Eddies, tides, and eddy–tide interactions of Australia’s North West Shelf*
- **Undergraduate Research Supervisor: Audrey Smart** 2024
Australian National University
 - Project: *Understanding the importance of model resolution for simulating tidal energy*

Research Cruises and Field Experience

- **CTD Watchstander**, GO-SHIP P15S Hydrographic Section (Leg 2) May–June 2016
R/V Investigator, South Pacific, Wellington, New Zealand–Lautoka, Fiji, 35 days
Monitoring ocean change and variability along 170°W, from the Antarctic ice edge to the equator

Professional Service

- **ACCESS-OM3 Development Collaborator** 2025–present
Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA)
- **Peer Reviewer** 2020–present
JPO, JAMES, JGR: Oceans, JFM, and GRL.
- **Group Meeting Coordinator** January 2024–December 2025
Climate and Fluid Physics, Research School of Earth Sciences, Australian National University
- **Discussion Leader** June 2024
Gordon Research Seminar on Ocean Mixing, South Hadley, Massachusetts, United States
- **Postdoctoral Fellow Selection Committee Member** March 2024
Australian National University

Science Outreach

- **Volunteer Science Demonstrator**, ANU Open Day 29 March 2025
Australian National University
 - Led an interactive “Icebergs in the Southern Ocean” experiment using a rotating tank and thermal camera, engaging prospective students and their families in discussions of ocean circulation and climate science.
- **Volunteer Instructor and Demonstrator**, IMAS Work Experience Program
Institute for Marine and Antarctic Studies, University of Tasmania June–July 2018
 - Designed and delivered a one-hour lecture, “How Do We Study the Oceans? Research in Physical Oceanography,” to two groups of high-school students from across Australia.
 - Led a rotating-tank experiment demonstrating ocean gyres and western boundary intensification.
- **Volunteer Educator**, Young Antarctic Scientists Program May and December 2018
Clarendon Vale Primary School, Tasmania
 - Delivered hands-on activities for Year 4 students introducing ocean layering and density, carbon dioxide chemistry, and drone-based environmental observations.